

Title

Cyber Security Toolkit

Project Name

Cyber Security Toolkit Using Java

Objective

The objective of this project is to develop a Java-based Cyber Security Toolkit that provides basic security utilities such as password strength checking, password generation, text encryption, text decryption, and SHA-256 hash generation. The project helps users understand fundamental cybersecurity concepts through a simple console application.

Problem Statement

Many users use weak passwords and are unaware of basic cybersecurity techniques. Sensitive information is often stored without encryption or secure hashing. This project provides simple tools that improve password security and demonstrate encryption and hashing techniques.

Research Findings

- Strong passwords reduce unauthorized access risks.
- Randomly generated passwords are more secure.
- Encryption protects confidential information.
- SHA-256 hashing securely protects data.
- Java provides built-in cryptographic libraries.

Methodology

1. Design a menu-driven application.
2. Accept user input.
3. Execute the selected module.
4. Display the output.
5. Return to the main menu or exit.

Tools Used

Programming Language: Java
IDE: Visual Studio Code
JDK 17 (or installed version)
GitHub
Windows/Linux

Implementation

The application includes the following modules:

- Password Strength Checker
- Password Generator
- Text Encryption (Caesar Cipher)
- Text Decryption
- SHA-256 Hash Generator

Screenshots

Insert screenshots of:

- Main Menu
- Password Generator Output
- Password Strength Checker Output
- Encryption Output
- Decryption Output
- SHA-256 Hash Output

Results

The project successfully performs password strength checking, secure password generation, text encryption, text decryption, and SHA-256 hash generation. The application demonstrates fundamental cybersecurity concepts effectively.

Challenges

- Understanding Java cryptography libraries.
- Implementing modular programming.
- Managing user input correctly.
- Testing different security functions.

Future Scope

Future enhancements may include a GUI, AES/RSA encryption, file encryption, password manager, database integration, and user authentication.

Conclusion

The Cyber Security Toolkit is a simple Java application demonstrating essential cybersecurity concepts. It provides practical tools for password security, encryption, decryption, and hashing while improving programming and cybersecurity knowledge.